

CS-1004

Object Oriented Programming

Assignment 04

Objectives:

-  Inheritance
-  Composition
-  Friend Function

Total Questions: 05

Total Marks: 100

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. There must be a block of comments at the start of every question's code by students; the block should contain brief description about functionality of code.
2. Comment on every function and about its functionality.
3. Mention comments where necessary such as comments with variables, loop, classes etc. to increase code understandability.
4. Use understandable names of variables.
5. Proper indentation of code is essential.
6. Write a code in C++ language.
7. Make a **Microsoft Word file** and paste all your C++ code with all possible screenshots of every task **output in Microsoft Word and submit the word file.**
8. First think about statement problems and then write/draw your logic on copy. After copy pencil work, code the problem statement on MS Studio C++ compiler. 10. At the end when you have done your tasks, attached C++ created files in MS word file and make your submission on Google Classroom. (Make sure your submission is completed).
9. Please submit your file in this format **22F_XXXX_A1_Sec.**
10. **Do not submit your assignment after the deadline. Late submission is not accepted.**
11. **Do not copy code from any source otherwise you will be penalized with negative marks or zero.**

Problem 1: | (Inheritance)

Design a class named **PersonData** with the following member variables:

- FirstName
- LastName
- Address

Write the appropriate **getter** and **setter** functions for these member variables.

Next, design a class named **CustomerData**, which is derived from the **PersonData** class. The CustomerData

Class should have the following member variables:

- CustomerNumber
- MailingList

The CustomerNumber variable will be used to hold a unique integer for each customer. The MailingList variable should be a bool. It will be set to true if the customer wishes to be on a mailing list, or false if the customer does not wish to be on a mailing list. Write appropriate getter and setter functions for these member variables. CustomerData class will have the

- InputCustomerData member function which will Input all the data for customer.
- DisplayCustomerData member function which will display all the data for customer.

Demonstrate an object of the CustomerData class in a simple program.

Use three file structure

Problem 2: | (Multi level Inheritance)

A retail store has a preferred customer plan where customers may earn discounts on all their purchases. The amount of a customer's discount is determined by the amount of the customer's cumulative purchases in the store.

- When a preferred customer spends \$500, he or she gets a 5% discount on all future purchases.
- When a preferred customer spends \$1,000, he or she gets a 6% discount on all future purchases.
- When a preferred customer spends \$1,500, he or she gets a 7% discount on all future purchases.
- When a preferred customer spends \$2,000 or more, he or she gets a 10% discount on all future purchases.

Design a class named PreferredCustomer, which is derived from the CustomerData class you created in **problem 1**. The PreferredCustomer class should have the following member variables:

- purchasesAmount (a double)
- discountLevel (a double)

The purchasesAmount variable holds the total of a customer's purchases to date. The discountLevel variable should be set to the correct discount percentage, according to the store's preferred customer plan. Write appropriate member functions for this class and demonstrate it in a simple program.

Use three file structure

Problem 3: | Composition

Create a class Time with following data members and member functions

```
public:
    Time();
    Time(int, int);
    void setTime(int, int);
    void getTime(int&, int&);
    void printTime();
private:
    int hr;
    int min;
```

Create a class Date with following data members and member functions

```
public:
    Date();
    Date(int, int, int);
    void setDate(int, int, int);
    void getDate(int&, int&, int&);
    void printDate();
private:
    int month;
    int day;
    int year;
```

Create a class Event with following data members and member functions

```
public:
    Event(int hours = 0, int minutes = 0, int m = 1,
int d = 1, int y = 1900, string name = "Christmas");
    void setEventData(int hours, int minutes, int m, int d, int y,
string name);
    void printEventData();
private:
    string eventName;
    Time eventTime;
    Date eventDay;
```

Write a Main and create events using the above classes to demonstrate composition.

Problem 4: | Death Relation

Implement the death relation using car as a whole class and engine, wheel, window, door and tires as a part classes.

(DEATH RELATION: The Whole class object will be created first and the constructor of those objects are called which are in the whole class and when whole class object is out of its scope than all the objects of whole and parts class are destroyed due to composition.)

Problem 5: | Friend Function

Class members can be accessed by other functions outside of the class own member functions called “friend” functions.

1. Create a class with the following member variables and functions:

- a) Initialize the member variables n1 and n2;
- b) Add two numbers and return the sum ($n1 + n2$)
- c) Subtract two numbers and return the difference ($n1 - n2$)
- d) Multiply two numbers and return the product ($n1 * n2$)
- e) Divide two numbers and return the dividend ($n1 / n2$). If n2 is zero, set n2 to 1.

2. The class has several friend functions:

- a) Average – pass the class object and return the average of n1 and n2
- b) Compare1 – pass the class object and return the Larger of n1 and n2. If n1 and n2 are equal zero signals that n1 and n2 are equal

c) Compare2 – pass the class object and return the Smaller of n1 and n2. If n1 and n2 are equal zero signals that n1 and n2 are equal

Output Example:

1 of 2 - Enter First numeric Value: 22.3

2 of 2 - Enter Second numeric Value: 44.5

First Value is 22.3

Second Value is 44.5

$22.3 + 44.5 = 66.8$

$22.3 - 44.5 = -22.2$

$22.3 \times 44.5 = 992.35$

$22.3 / 44.5 = 0.501124$

Average of 66.8 is 33.4

The Larger of 22.3 and 44.5 is 44.5

The Smaller of 22.3 and 44.5 is 22.3

Run this Again (Y or N):